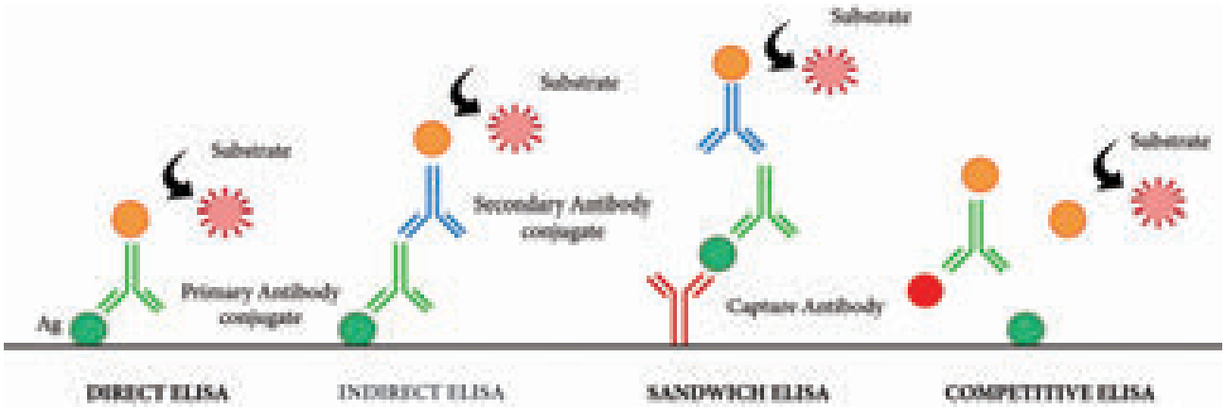


Figure 11.5: Types of ELISA

Types

There are **four kinds** of ELISA assay tests. They are: Direct ELISA, Indirect ELISA, Sandwich ELISA and Competitive ELISA (Figure 11.5).

i. Direct ELISA

An antigen is immobilized in the well of an ELISA plate. The antigen is then detected by an antibody directly conjugated to an enzyme such as HRP. Direct ELISA detection is much faster than other ELISA techniques as fewer steps are required. The assay is also less prone to error since fewer reagents and steps are needed, i.e. no potentially cross-reacting secondary antibody needed. Finally, the direct ELISA technique is typically used when the immune response to an antigen needs to be analyzed.

ii. Indirect ELISA

Indirect ELISA is used to detect antibody. A known antigen is coated on the micro titre plate. If the patient's serum contains antibody specific to the antigen, the antibody will bind to the antigen. After incubation the wells are washed and the enzyme labelled with anti Human Gamma Globulin (HGG) is added to the well. Anti-HGG can react with antigen antibody complex. The substrate for the enzyme is added finally which is hydrolysed by the enzyme which develops a colour.

iii. Sandwich ELISA

Sandwich ELISA is used to detect antigen. A known antibody is coated on the micro titre plate. A test antigen is added to each well and allowed to react with the bound antibody.

If the patient's serum contains antigen specific to the antibody, the antigen will bind to the antibody. Specifically bound antigen and antibody will remain in the wells even after washing. The second antibody is added and allowed to react with bound antigen. Substrate is added to measure colour reaction.

iv. Competitive ELISA

It is used for the detection of antigens. Antibody is first incubated with a sample-containing antigen. The antigen and antibody complex is added to the antigen coated microtitre well. If more antigen present in the sample, the less free antibody will be available to bind to the antigen coated well. Addition of an enzyme conjugated secondary antibody specific to the primary antibody can be used to determine the amount of primary antibody bound to the well. It is a quantitative test for the antigen detection.

TN_GOVT_XII_Micro_Biology_CH11.indd 233 04-03-2019 12:32:00